1. Initialization

Initialize class means μ = [0.5, 0.5, 0.5]
Initialize variances σ² = [0.1, 0.1, 0.1]
Initialize class weights W = [1, 1, 1]
Initialize tumor boost β = 1.0

Classes:
1: Background
2: Liver
3: Tumor

2. Training Loop
for epoch = 1 : MaxEpoch do

3. Model Evaluation

1. Train ResUNet using current class weights W

2. Compute validation metrics:
Dice_liver | Dice_tumor | Precision_liver

3. Compute composite reward r:
r = 0.3·Dice_liver + 0.5·Dice_tumor + 0.2·Precision_liver

4. Update tumor boost β
based on moving average of Dice_tumor

4. CLA Update (for each class c {1,2,3})

Update mean μ_c and variance σ²_c
using reward signal (Eqs. 14–17)

μ_c(t+1) = μ_c(t) + α r (x_c − μ_c(t))
σ²_c(t+1) = σ²_c(t) + α r [(x_c − μ_c(t))² − σ²_c(t)]

Compute Gaussian probability
(Eq. 12)

p_c = (1 / √(2πσ²_c)) · exp(−(x_c − μ_c)² / (2σ²_c))

Update class weight w_c
(Eq. 13)

w_c = (p_c / Σ p_j) · β_c

Apply tumor boost for class 3

β_c = β if c = 3
β_c = 1 otherwise

5. Weight Constraints
W = Clip(W, 1.0, 8.0)
i.e., 1.0 ≤ w_c ≤ 8.0 c {1,2,3}

Next epoch?

Yes

No

6. Output
• Predicted masks (Liver & Tumor)
• Final evaluation metrics
(Dice, IoU, Precision, Recall,
VOE, RVD, ASSD, MSD, ICC)